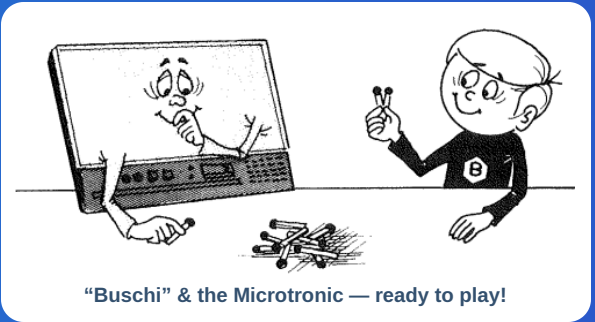


A REAL COMPUTER FROM 1981 — AND THE
SECRET INSIDE EVERY COMPUTER & PHONE

MEET THE MICROTRONIC!

It's a tiny computer you command **by hand** — no mouse, no apps, just a keypad and a glowing display. You type in instructions one at a time... and YOU become the programmer! 🤖



"Buschi" & the Microtronic — ready to play!

4

bits — it thinks in tiny numbers

16

little number boxes (registers)

3

digits = one instruction

1981

made in Germany by Busch

💡 BINARY — just 0 and 1

Inside every computer are millions of tiny **switches**. A switch is either **ON = 1** or **OFF = 0**. That's the whole alphabet computers use!



Four switches spell **0101** — that means **5**!

0000 = 0 0001 = 1 0010 = 2 0011 = 3 0100 = 4
0101 = 5 0110 = 6 0111 = 7 1000 = 8 1001 = 9

Try it: can you count to 5 on one hand, using each finger as a switch? 🙌

1234 HEX — a shortcut for 0–15

Four switches make **16** different values (0–15). Writing 1111 is long, so we give each value **one symbol** — after 9 we keep going with letters!

10	11	12	13	14	15
A	B	C	D	E	F

So **F** means **15**! This is called **hexadecimal** ("hex"). The Microtronic shows *everything* in hex — and each number box holds one hex digit, 0 to F.

💻 WRITE YOUR FIRST PROGRAM!

Every Microtronic instruction is just **3 hex digits**. This is **machine code** — the real language your phone, game console and every computer secretly run, billions of times a second. Try this:

1 5 1 → put the number **5** into Box 1
5 3 1 → add **3** to Box 1

🎉 **Box 1 now holds 5 + 3 = 8 — you just programmed a computer!**

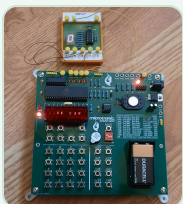
⚡ If a box ever goes past **F (15)**, the **Carry light** blinks — that's how computers count to really big numbers, one nibble at a time.

🌐 WHY IT MATTERS

The Microtronic is slow enough that you can **see every step** a computer takes. Learn it, and you understand what's *really* happening inside **all** computers — phones, laptops, robots, spaceships.



It's **4-bit** and does only a few instructions per second... but it works exactly like the big machines, just one tiny move at a time.



🔧 Want to become a programmer? Build your own!

Ask a grown-up to help you build the **Microtronic Phoenix** — a modern kit that even runs the **real 1981 program chip**. It's free and open-source:

github.com/lambda-mikel/microtronic-phoenix

Need parts or help assembling it? Have a parent email me:

miacwess@gmail.com ✉

Buschi says: come find me at the table — let's play a game of NIM, and I'll show you the little computer up close! · Busch Microtronic

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